

Lab Experiment No. 03

Design an IPv4 addressing scheme and implement subnetting in a small network using Cisco Packet Tracer

Objective:

To design an IPv4 addressing scheme and implement subnetting in a small network using Cisco Packet Tracer on the example 192.168.100.0/25.

Problem Statement:

You are required to design an IPv4 addressing scheme and implement subnetting for a small network using the IP address block 192.168.100.0/25. The network should be divided into subnets and the addressing scheme must be assigned to hosts and network devices using Cisco Packet Tracer.

Theory:

Subnetting is the process of dividing a single network into multiple smaller networks (subnets). This improves network management, security, and efficient use of IP addresses.

The given network address is 192.168.100.0/25. The /25 prefix means that the first 25 bits are reserved for the network portion, leaving 7 bits for host addresses.

Number of hosts per subnet = $2^7 - 2 = 126$ usable addresses.

Therefore, the given address space (192.168.10.0/25) provides one subnet with the following details:

- Network Address: 192.168.100.0
- First Usable Address: 192.168.100.1
- Last Usable Address: 192.168.100.126
- Broadcast Address: 192.168.100.127

Subnetting with Classless Addressing

- For Example: Lets take a class – C address as:

Lets take class – C Address as = 192.168.100.0/25 (bit borrowed from host id : 1)

No. of Subnets = $2^1 = 2$ No. of Hosts/Subnet = $2^7 - 2 = 128 - 2 = 126$

Subnets Mask for both subnets : 255.255.255.128



Subnetting helps in logical distribution of IP addresses to different devices such as PCs, switches, and routers.

Solution:

Step 1: Open Cisco Packet Tracer and create a small network topology with two PCs connected via a switch and a router.

Step 2: Assign IP addresses from the subnet 192.168.100.0/25 to each device:

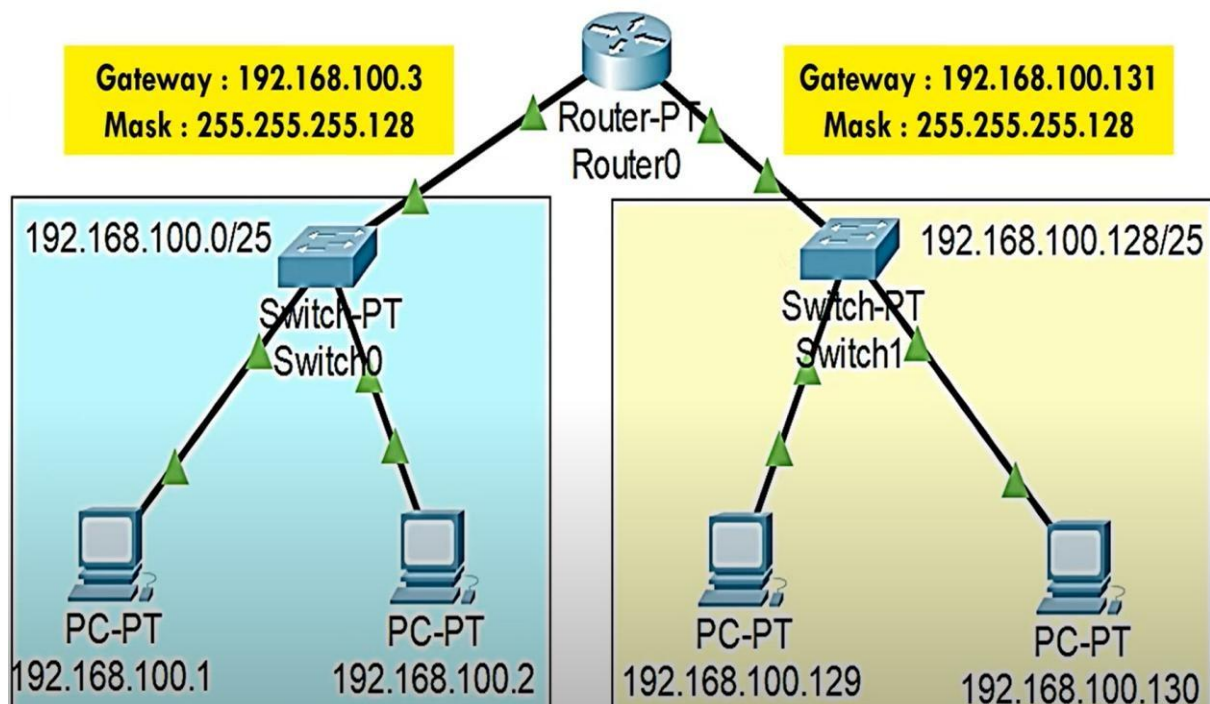
- PC1: 192.168.100.10/25, Gateway: 192.168.100.1
- PC2: 192.168.10.20/25, Gateway: 192.168.100.1
- Router (Interface connected to switch): 192.168.100.1/25

192.168.100.0/25			
Subnet Address	First Host Address	Last Host Address	Broadcast Address
192.168.100.0/25	192.168.100.1	192.168.100.126	192.168.100.127
192.168.100.128/25	192.168.100.129	192.168.100.254	192.168.100.255

Step 3: Configure the devices in Packet Tracer with the assigned IPs.

Step 4: Test connectivity by pinging from one PC to another PC.

The diagram below represents the logical setup of the network:



[Network Diagram: Router connected to Switch, Switch connected to PC]

Output:

Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

Logical Physical x: 400, y: 184 Root

Time: 25:18:13

Scenario 0

New Delete

Toggle PDU List Window

File	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
Successful	PC0	Router1	ICMP		0.000	N	0	(e...)		(delete)
Successful	PC0	Router1	ICMP		0.000	N	1	(e...)		(delete)
Successful	PC1	Router1	ICMP		0.000	N	2	(e...)		(delete)
Successful	PC3	Router1	ICMP		0.000	N	3	(e...)		(delete)

Activate Window (delete)

Go to Settings to delete (delete)

Conclusion:

The IPv4 addressing scheme using the 192.168.100.0/25 network was successfully implemented in Cisco Packet Tracer. Both PCs were able to communicate with each other using the assigned subnet, demonstrating correct subnetting and addressing.